

1. Part 1: Differential Amplifier Design

Design Specifications

We want to design a resistive-loaded differential amplifier with the following specs.

Parameter	Value
Supply (V_{DD})	1.8V
Current seed (Ref current)	$20\mu\text{A}$
Bias current (I_{SS})	$40\mu\text{A}$
Differential gain	8
CM output level	$V_{DD}/3$
Load capacitance	1pF

Note: the bias current is split between two transistors; each transistor gets $I_D = 20\mu\text{A}$.

Input Stage Type

Q: Since the output level is closer to the ground rail, why do we choose a PMOS input stage (with the PMOS not placed in a dedicated well, to save area)?

A PMOS input stage is required because it can remain safely in saturation even when its input and output nodes swing close to the ground rail so that $V_{DG} < V_{th}$

R_D Selection

Q: Choose R_D to meet the CM output level spec. Show your calculation.

$$R_D = V_{DD}/3I_D = 30k\Omega$$

LUT Settings

LUT: pmos_03v3

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 20u

gm/ID: $8/(0.91 \cdot 0.6)$

ro: 6/ID

VDS: 0.9

VSB: 0.3

Stack: 1

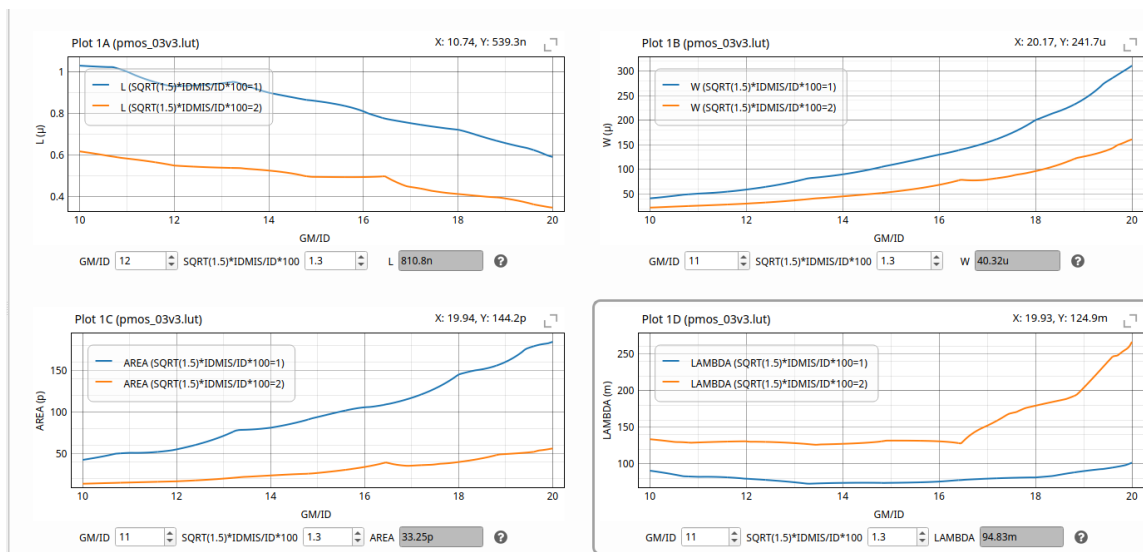
Results:

	Name	TT-27.0
1	ID	20u
2	IG	N/A
3	L	350n
4	W	30u
5	VDB	1.2
6	VGB	1.241

Figure 1: Sizing of Input Pair

Tail Current Source Design

Parameter	Value
Input current	$20\mu A$
Percent mismatch: $\sigma(I_{out})/I_{out}$	$\leq 2\%$
Lambda	0.1
Compliance voltage	$\leq 200mV$
Area	Minimize

Figure 2: Sizing of Current Mirror at $V_{DS} = 0.9$

Mismatch vs Lambda Trade-off

Q: In order to satisfy both the mismatch and lambda requirements, which design curve will you pick, the 1% or the 2% mismatch?

I will tend toward the 2% mismatch since it requires lesser area.

CM Input Level Range

Q: Calculate the min and max CM input levels. Is the previously selected CM input level in the valid range?

2. Part 2: Differential Amplifier Simulation

2.1 1. OP Simulation

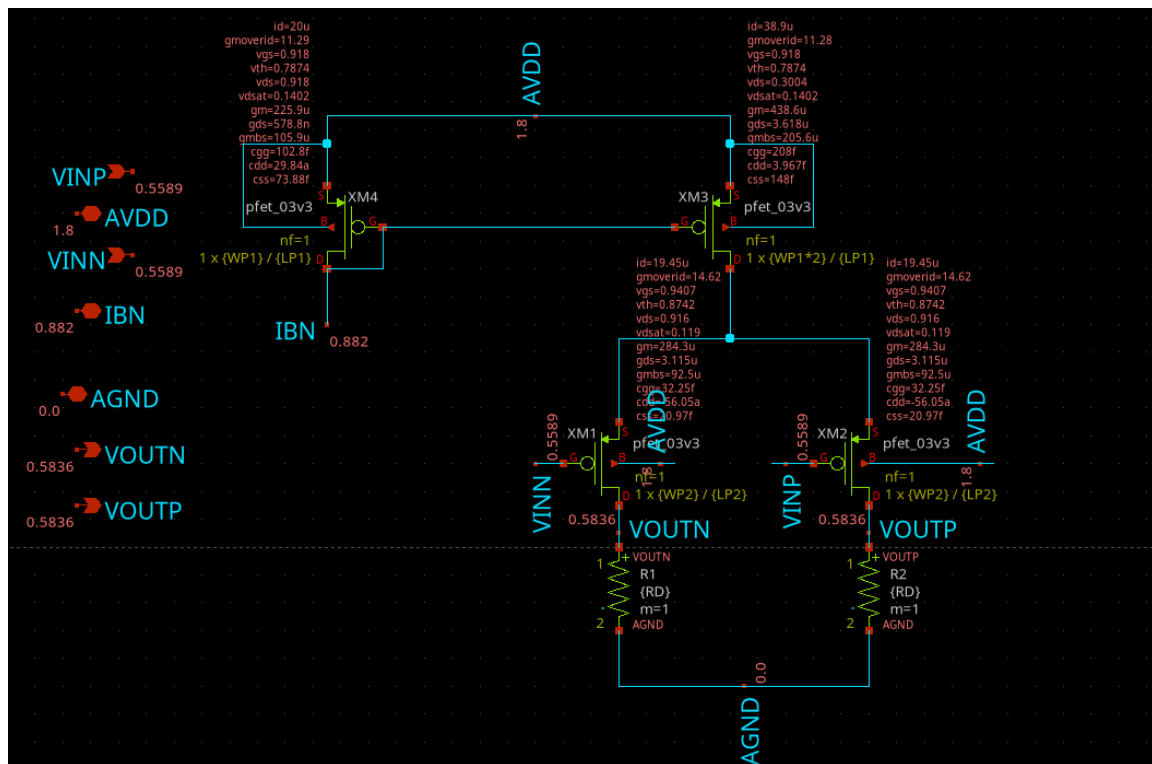


Figure 3: DC Operating Points

Parameter	M1	M2	M3	M4
$V^*(mV) = 2/(g_m/I_D)$	136.79	136.79	177.3	177.14
Region	2	2	2	2

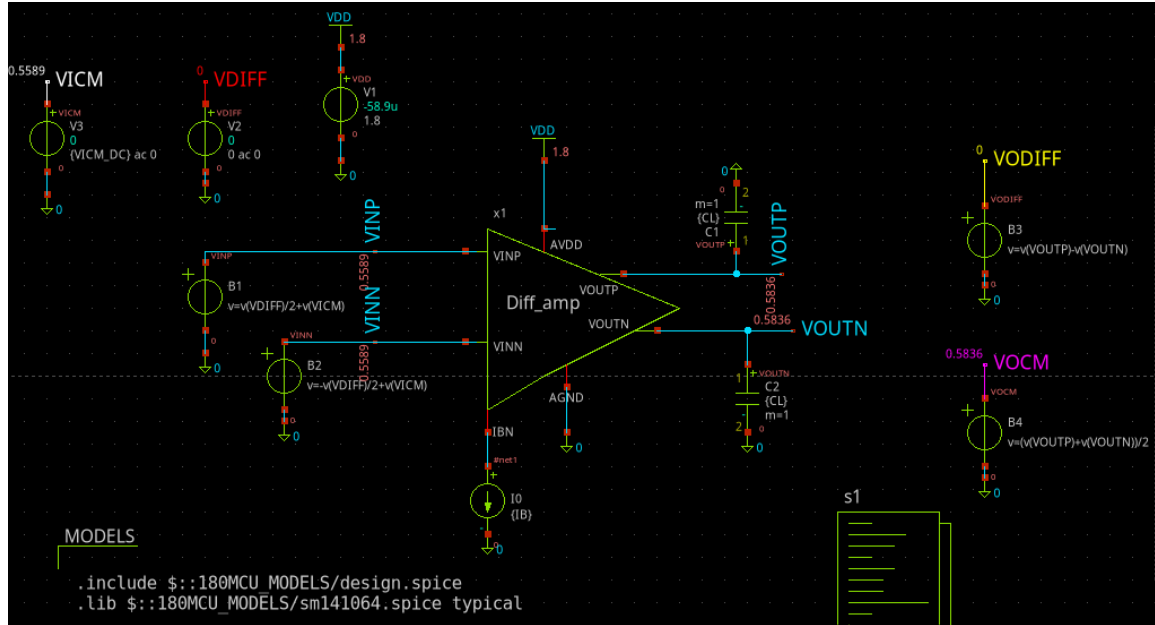


Figure 4: Testbench

2.2 2. Diff Small-Signal AC Characteristics

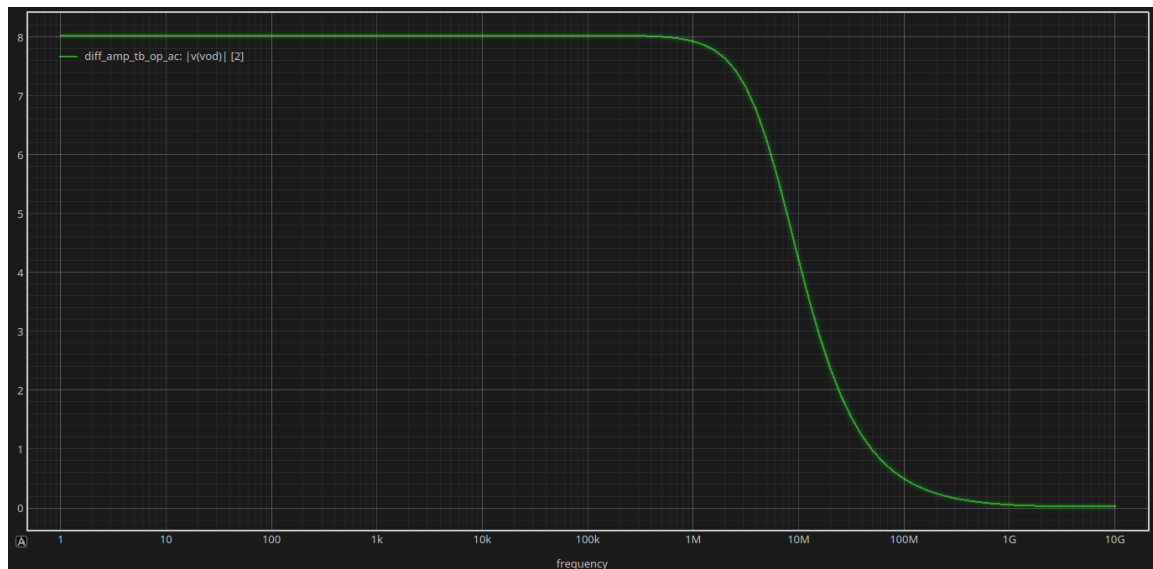


Figure 5: Bode Magnitude plot for small signal differential gain

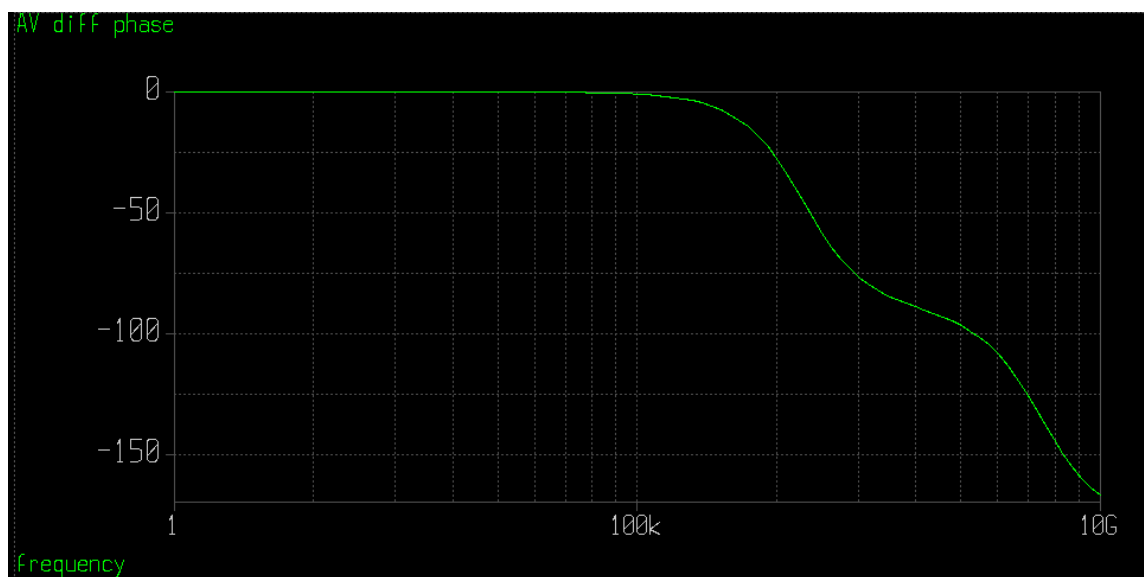


Figure 6: Phase of Differential small signal gain

Diff Gain and Bandwidth Comparison

Q: Compare the DC diff gain and BW with hand analysis in a table.

$$BW = \frac{1}{2\pi(27.43 \text{ k}\Omega)(1 \text{ pF})}$$

$$BW = \frac{1}{2\pi(27.43 \times 10^3)(1 \times 10^{-12})}$$

$$BW \approx 5.80 \text{ MHz}$$

Metric	Hand Analysis	Simulation
$A_{v,diff}$ (dB)	18.06	17.84
BW (Hz)	5.8M	5.68M

2.3 3. CM Small-Signal AC Characteristics

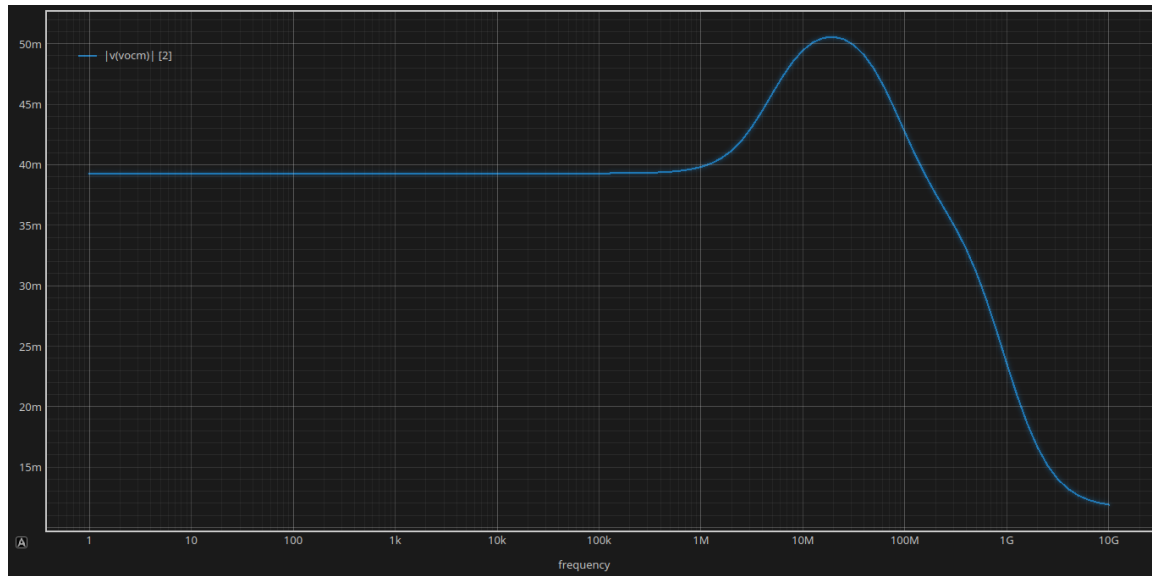


Figure 7: Small Signal Common Mode gain vs frequency

Hand Analysis

$$A_{cm} = \frac{-g_{m3}(R_D \parallel r_{o2})}{1 + 2g_{m2}r_{o3}}$$

$$A_{cm} = \frac{-(284.3 \mu\text{S})(30 \text{ k}\Omega \parallel 321.03 \text{ k}\Omega)}{1 + 2(284.3 \mu\text{S})(276.4 \text{ k}\Omega)}$$

$$A_{cm} = \frac{-7.79}{1 + 157.16}$$

$$A_{cm} \approx -0.0492 \text{ V/V} \quad (-26.16 \text{ dB})$$

CM Gain Comparison

Q: Compare the DC CM gain with hand analysis in a table. Is it smaller than “1”? Why?

Metric	Hand Analysis	Simulation
$A_{v,CM}$ (dB)	-26.16	-26.02

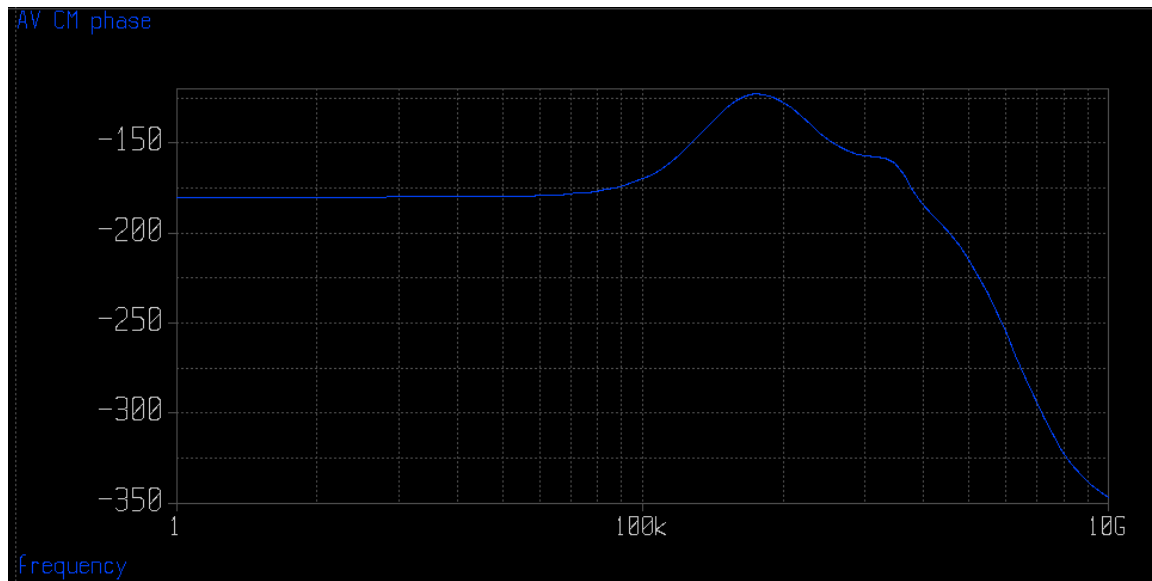


Figure 8: Small Signal Common Mode gain Phase vs frequency

CM Gain Frequency Variation

Q: Justify the variation of A_{vcm} vs frequency.

The initial rise in common-mode gain is caused by the tail node's parasitic capacitance acting as an AC short, which destroys the source degeneration that normally suppresses common-mode signals. The subsequent high-frequency roll-off occurs because the output capacitance shorts RD

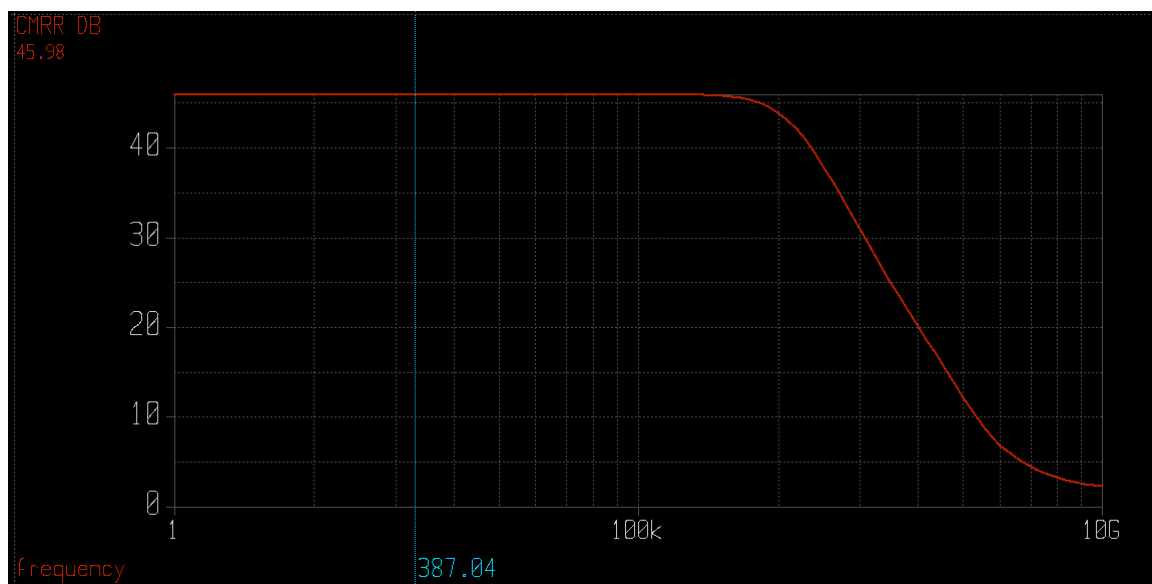


Figure 9: CMRR (dB) vs frequency

CMRR Comparison

Q: Plot A_{vd}/A_{vcm} in dB. Compare A_{vd}/A_{vcm} @ DC with hand analysis in a table.

Metric	Hand Analysis	Simulation
CMRR (dB)	44.22	45.98

CMRR Frequency Variation

Q: Justify the variation of A_{vd}/A_{vcm} with frequency.

The ratio A_{vd}/A_{vcm} degrades as frequency increases primarily because the parasitic capacitance at the tail node acts as an AC short that destroys common-mode suppression, causing A_{vcm} to spike while A_{vd} simultaneously rolls off due to output capacitance.

2.4 4. Diff Large-Signal Characteristics

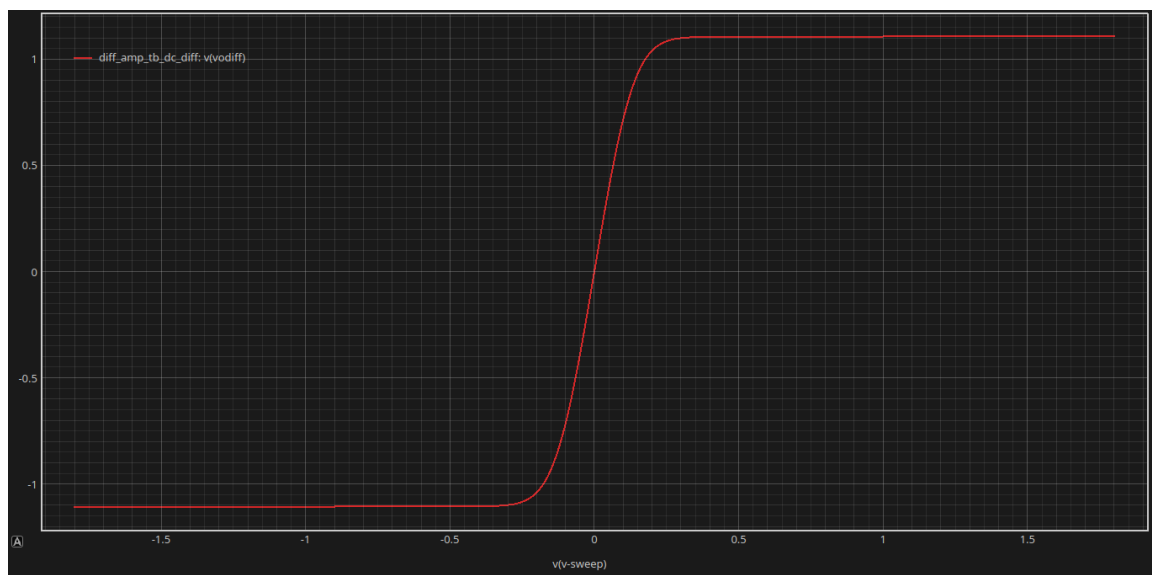


Figure 10: Differential output voltage vs Input differential

Extreme Values Comparison

Q: Compare the extreme values (V_{max} , V_{min} of V_{ODIFF}) with hand analysis in a table.

Metric	Hand Analysis	Simulation
V_{odmax}	1.2	1.18
V_{odmin}	-1.2	-1.18

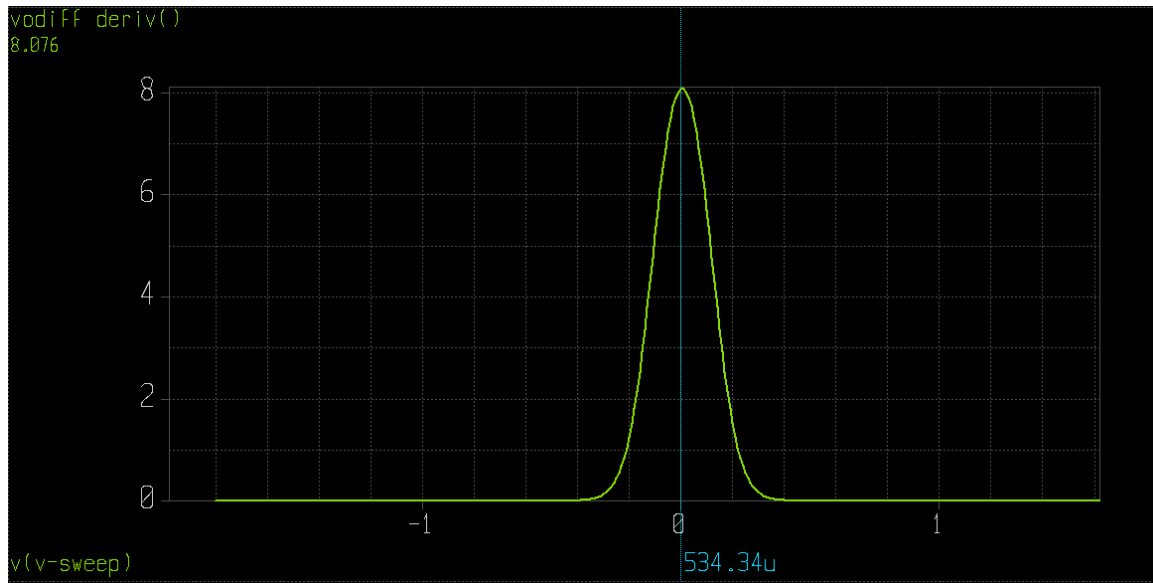


Figure 11: Derivative of output differential voltage vs input

Peak Derivative

Q: What is the peak value of $\text{deriv}(V_{ODIFF})$? Why?

It is the same value of the small signal differential gain at $v_{id}=0$

2.5 5. CM Large-Signal Characteristics

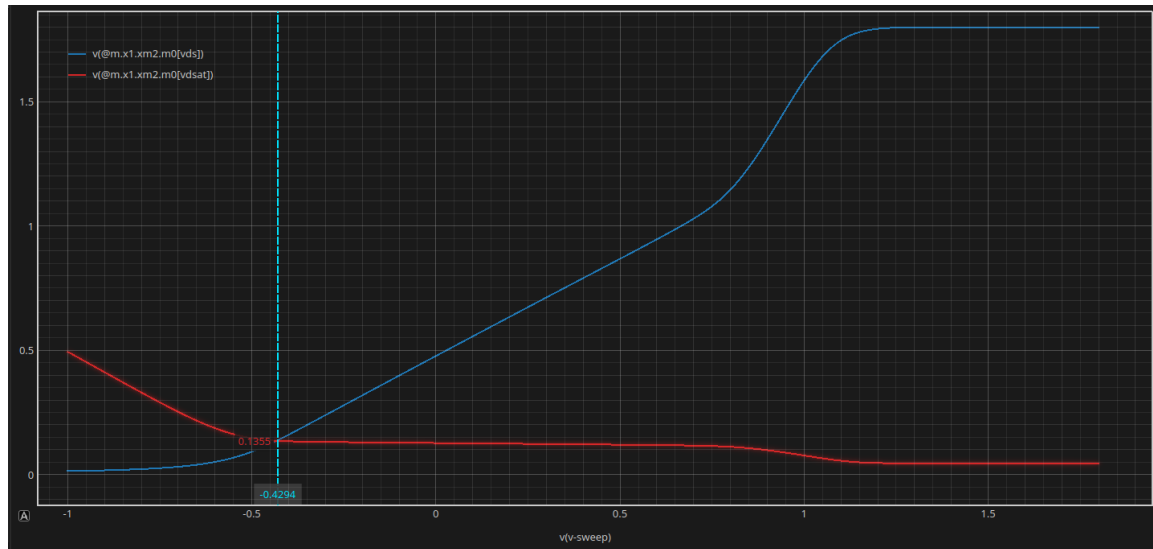


Figure 12: VDS of input pair vs differential input voltage

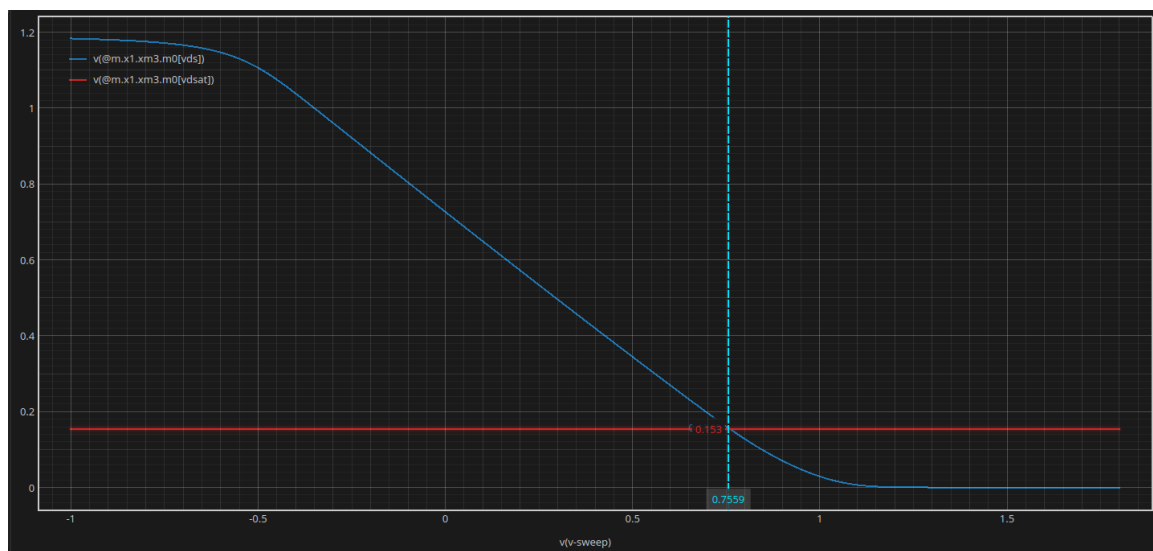


Figure 13: VDS of tail current source vs differential input voltage

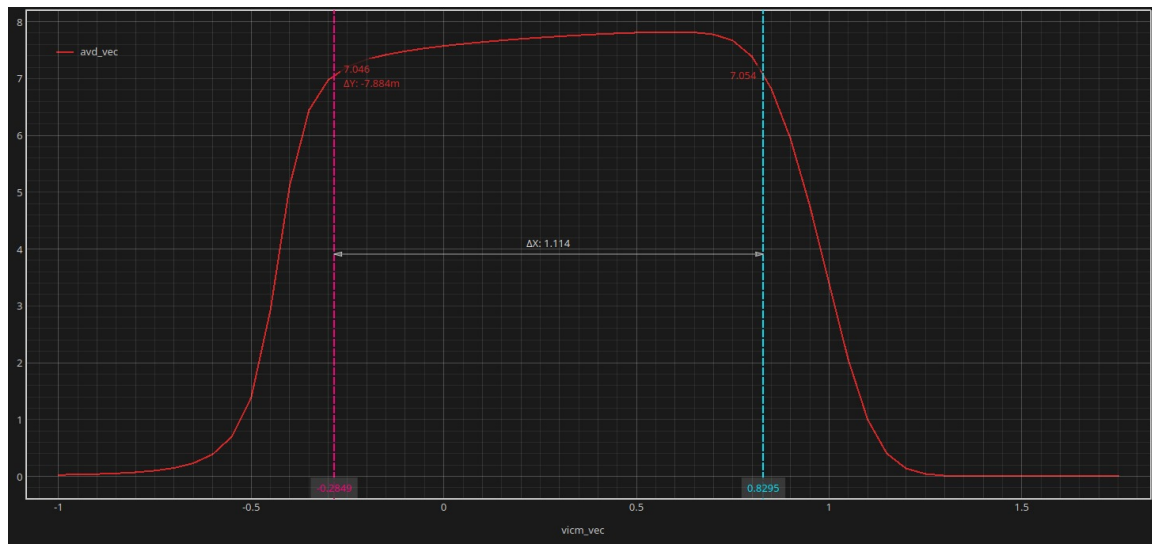


Figure 14: Differential Small signal gain vs Common Mode Input

CMIR Comparison

Q: Find the CM input range (CMIR). Compare with hand analysis in a table.

Metric	Hand	Simulation
$V_{ICM,min}$	-0.2	-0.43
$V_{ICM,max}$	0.66	0.76

Min V_{ICM} Discrepancy

Q: The min V_{ICM} is not as expected, why?

Because our hand analysis involves using V^* which is higher than v_{dsat} used by the simulator. Furthermore, We ignore V_{SB} in our calculations.

